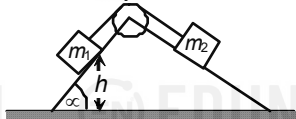


SUBJECTIVE TYPE

1. Two blocks of masses m_1 and m_2 are connected by a massless, inextensible, cord that is passed over a massless pulley A . The blocks slide along the smooth sides of a right angled wedge of mass m , which rests on a smooth horizontal plane. Find the distance covered by the wedge on the horizontal plane till the mass m_1 is lowered by the vertical distance h .



Solution In the frame of wedge, let s be the distance through which the blocks slide on the wedge.

Then, $x = h / \sin \alpha$

If the wedge slides by horizontal distance x towards right, then the horizontal distance through which the blocks m_1 and m_2 slide in the ground frame are respectively

$$x_1 = x - s \cos \alpha = x - h \cot \alpha \quad \text{and}$$

$$x_2 = x - s \cos (90^\circ - \alpha) = x - h$$

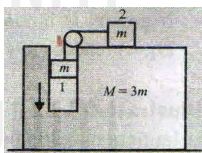
Since no external force acts on the system in horizontal direction and the system was initially at rest, the centre of mass does not move in the horizontal direction. So,

$$\sum m_i x_i = 0 \Rightarrow mx + m_1 x_1 + m_2 x_2 = 0$$

$$\Rightarrow mx + m_1 (x - h \cot \alpha) + m_2 (x - h) = 0$$

$$\therefore x = \frac{m_1 h \cot \alpha + m_2 h}{m + m_1 + m_2}$$

2. A wedge of mass $M = 3m$ and having a vertical slot in it is placed on a horizontal surface. Two blocks each of mass m are arranged as shown in the figure. The system is released from rest. All surface are smooth. Calculate the speed of the wedge when block 1 comes down a distance h .



Solution Let u be the vertical velocity of block 1 when it comes down a distance h . Let v be the velocity of wedge towards right at this instant. Then, the velocity of block 2 is $v - u$ towards right and of block 1 is v towards right. Since, there is no external force on the system in horizontal direction, momentum in the horizontal direction is conserved.

$$\Rightarrow (3m)v + mv + m(v - u) = 0 \Rightarrow u = 5v$$

Since, all surfaces are smooth, mechanical energy is conserved. So, Gain in KE = Loss in PE

$$\Rightarrow \frac{1}{2}(3m)v^2 + \frac{1}{2}m(u^2 + v^2) + \frac{1}{2}m(v - u)^2 = mgh$$

On putting $u = 5v$ in this equation, we get

$$v = \sqrt{2gh/45}$$

3. Two balls of mass $m_1 = 100 \text{ g}$ and $m_2 = 300 \text{ g}$ are suspended from point A by two equal inextensible threads, each of length $\ell = 32/35 \text{ m}$.

Ball of mass m_1 is drawn aside and held at the same level as A but at a distance $\sqrt{3}\ell/2$ from A , as shown in fig. When ball m_1 is released, it collides elastically with the stationary ball of mass m_2 . Calculate

(a) velocity u_1 with which the ball of mass m_1 collides with the other ball

(b) maximum rise of the ball of mass m_2

(Take $g = 10 \text{ ms}^{-2}$)

Solution:

$$\text{Here, } \sin \theta = \frac{\sqrt{3}\ell/2}{\ell} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = 60^\circ$$

The ball of mass m_1 will first fall vertically a distance of

$$\ell \cos \theta = \ell/2$$

Its velocity on reaching point

$$C \text{ is } v = \sqrt{2g\ell/2} = \sqrt{g\ell}$$

Its velocity at C in tangential direction will not change and will be equal to $v \sin \theta = \sqrt{3g\ell}/2$

(a) On reaching lowermost point, its velocity u_1 can be found by applying conservation of mechanical energy between C and lowermost position

Gain in KE = Loss in PE

$$\Rightarrow \frac{1}{2}m[u_1^2 - (v \sin \theta)^2] = mg\ell(1 - \cos \theta)$$

$$\Rightarrow u_1^2 = 2g\ell \times \frac{1}{2} + \left(\frac{\sqrt{3g\ell}}{2}\right)^2 = \frac{7g\ell}{4} = \frac{7 \times 10 \times 32/35}{4} = 16$$

$$\therefore u_1 = 4 \text{ m/s}$$

(b) Applying momentum conservation, we have

$$m_1 u_1 + 0 = m_1 v_1 + m_2 v_2$$

$$\Rightarrow 0.1 \times 4 = 0.1v_1 + 0.3v_2$$

$$\Rightarrow v_1 + 3v_2 = 4 \quad \dots(1)$$

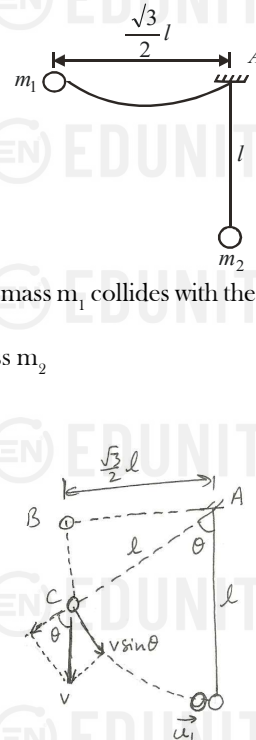
Since, collision is elastic,

$$v_2 - v_1 = u_1 - u_2 = 4 \quad \dots(2)$$

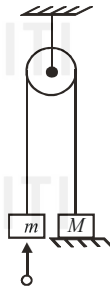
From (1) and (2), we get $v_2 = 2 \text{ m/s}$

Therefore, maximum rise of ball of mass m_2 is

$$h = \frac{v_2^2}{2g} = \frac{2^2}{2 \times 10} = 0.2 \text{ m}$$



4. A light flexible thread passes over a small, frictionless pulley. Two blocks of mass $m = 1$ kg and $M = 3$ kg are attached with the thread as shown in fig. Heavier block rests on a slab. A bullet of mass $m_0 = 1$ kg, moving upwards with velocity $v_0 = 10 \text{ ms}^{-1}$, collides with the hanging block at time $t = 0$. Calculate the maximum height ascended by M when it is jerked into motion



(Take $g = 10 \text{ ms}^{-2}$)

- (a) if the bullet gets stuck in the hanging block
(b) if the bullet collides with the hanging block elastically

Solution:

- (a) From conservation of momentum, the speed of block of mass m (with bullet) after the bullet strikes is

$$v = \frac{m_0 v_0}{m + m_0}$$

Block of mass M is jerked into motion when mass m (with bullet) returns back to its original position with momentum $(m + m_0) v = m_0 v_0$

The velocity of mass M at this instant is

$$v_1 = \frac{m_0 v_0}{M + m + m_0} = \frac{1 \times 10}{3 + 1 + 1} = 2 \text{ m/s}$$

Its acceleration is

$$a_1 = \frac{(m + m_0) - M}{m + m_0 + M} g = \frac{(1 + 1) - 3}{1 + 1 + 3} \times 10 = -2 \text{ m/s}^2$$

Maximum height (h_1) ascended is given by

$$h_1 = \frac{0^2 - v_1^2}{2a_1} = \frac{-2^2}{2 \times (-2)} = 1 \text{ m}$$

- (b) When the bullet strikes elastically, velocities interchange and the velocity of block of mass m is v_0 . Block of mass M is jerked into motion when mass m returns back to its original position with momentum mv_0 .

The velocity of mass M at this instant is

$$v_2 = \frac{mv_0}{m + M} + \frac{1 \times 10}{1 + 3} = 2.5 \text{ m/s}$$

Its acceleration is

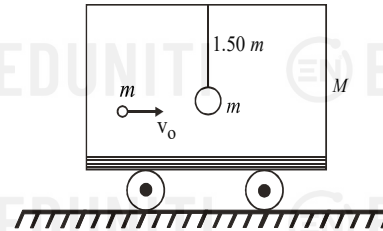
$$a_2 = \frac{m - M}{m + M} g = \frac{1 - 3}{1 + 3} \times 10 = -5 \text{ m/s}^2$$

Maximum height (h_2) ascended is given by

$$h_2 = \frac{0^2 - v_2^2}{2a_2} = \frac{-2.5^2}{2 \times (-5)} = 0.625 \text{ m}$$

5. A ball of mass $m = 1$ kg is hung vertically by thread of length $l = 1.50$ m. Upper end of the thread is attached to the ceiling of a trolley of mass $M = 4$ kg. Initially, trolley is stationary and is free to move along horizontal rails without friction. A bullet of mass $m = 1$ kg, moving horizontally with velocity $v_0 = 6 \text{ ms}^{-1}$, collides with the ball and gets stuck with it. As a result, thread starts to deflect towards right. Calculate its maximum deflection with the vertical.

(Take $g = 10 \text{ ms}^{-2}$)



Solution: From momentum conservation, the velocity of the ball (with bullet) after the bullet collides is

$$v'_0 = \frac{mv_0}{2m} = \frac{v_0}{2}$$

At the time of maximum deflection of thread, the velocity of ball and trolley is same. Let this be v .

From momentum conservation again, we have

$$2m \times \frac{v_0}{2} = (2m + M) v$$

$$\Rightarrow v = \frac{mv_0}{2m + M} = \frac{1 \times 6}{2 \times 1 + 4} = 1 \text{ m/s}$$

Applying conservation of mechanical energy, we have

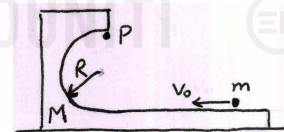
Gain in PE = Loss in KE

$$\Rightarrow 2mg\ell(1 - \cos\theta) = \frac{1}{2}(2m)\left(\frac{v_0}{2}\right)^2 - \frac{1}{2}(2m + M)v^2$$

$$\Rightarrow 2 \times 1 \times 10 \times 1.5(1 - \cos\theta) = 1 \times 3^2 - \frac{1}{2} \times 6 \times 1^2$$

$$\Rightarrow \cos\theta = 0.8 \quad \therefore \theta = 37^\circ$$

6. A small ball of mass m is projected with a minimum horizontal velocity v_0 on a block of mass M so that it will reach the highest point P of the block. If all surfaces are smooth, find the value of v_0



Solution For v_0 to be minimum, the speed of ball with respect to block at the highest point P is $\sqrt{gR}$. If at this instant, speed of block is v towards left, then the speed of ball with respect to ground is $v - \sqrt{gR}$ towards left.

As there is no external force acting on the system in horizontal direction, the linear momentum in horizontal direction is conserved.

$$\Rightarrow mv_0 = Mv + m(v - \sqrt{gR}) \Rightarrow v = \frac{m(v_0 + \sqrt{gR})}{M + m}$$

By conservation of mechanical energy,

$$\Delta K + \Delta U = 0$$

$$\Rightarrow \frac{1}{2}Mv^2 + \frac{1}{2}m(v - \sqrt{gR})^2 - \frac{1}{2}mv_0^2 + mg(2R) = 0$$

$$\Rightarrow (M + m)v^2 - 2m\sqrt{gR}v + mgR - mv_0^2 + 4mgR = 0$$

$$\Rightarrow \frac{M + m}{m} \left[\frac{m(v_0 + \sqrt{gR})}{M + m} \right]^2 - 2\sqrt{gR} \left[\frac{m(v_0 + \sqrt{gR})}{M + m} \right] - v_0^2 + 5gR = 0$$

$$\Rightarrow \frac{m}{M + m} (v_0^2 + 2\sqrt{gR}v_0 + gR) - \frac{m}{M + m} (2\sqrt{gR}v_0 + 2gR) - v_0^2 + 5gR = 0$$

$$\Rightarrow v_0^2 \left(1 - \frac{m}{M + m} \right) = gR \left(5 - \frac{m}{M + m} \right)$$

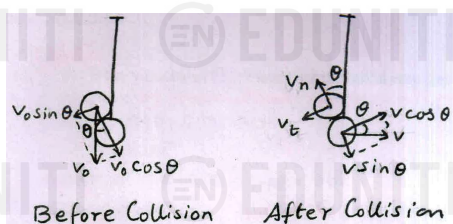
$$\Rightarrow Mv_0^2 = gR(5M + 4m)$$

$$\therefore v_0 = \sqrt{gR \left(5 + \frac{4m}{M} \right)}$$

7. A small steel ball A is suspended by an inextensible thread of length $\ell = 1.5\text{ m}$ from O. Another identical ball is thrown vertically downwards such that its surface remains just in contact with thread during downward motion and collides elastically with the suspended ball. If the suspended ball just completes vertical circle after collision, calculate the velocity of the falling ball just before collision.

(Take $g = 10\text{ ms}^{-2}$)

Solution:



It can be seen that $\theta = 30^\circ$.

The velocity along the tangent remains same.

$$\Rightarrow v_t = v_0 \sin \theta$$

Since collision is elastic,

$$v_2 - v_1 = u_1 - u_2$$

$$\Rightarrow v \sin \theta - (-v_n) = v_0 \cos \theta - 0$$

$$\Rightarrow v_n = v_0 \cos \theta - v \sin \theta$$

Since there is no external force in horizontal direction, the

momentum of the system of two balls along horizontal is conserved.

$$\Rightarrow mv = m(v_t \cos \theta + v_n \sin \theta)$$

$$\Rightarrow v = v_0 \sin \theta \cos \theta + (v_0 \cos \theta - v \sin \theta) \sin \theta$$

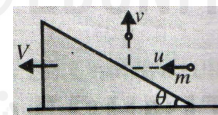
$$\Rightarrow v = \frac{2v_0 \sin \theta \cos \theta}{1 + \sin^2 \theta} = \frac{v_0 \sin 2\theta}{1 + \sin^2 \theta}$$

For the ball to just complete the circle, $v = \sqrt{5gl}$

$$\Rightarrow \frac{v_0 \sin 2\theta}{1 + \sin^2 \theta} = \sqrt{5gl}$$

$$\therefore v_0 = \frac{\sqrt{5gl}(1 + \sin^2 \theta)}{\sin 2\theta} = \frac{\sqrt{5 \times 10 \times 1.5}(1 + \sin^2 30^\circ)}{\sin 60^\circ} = 12.5\text{ m/s}$$

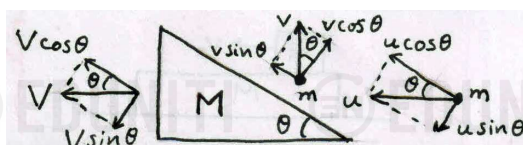
8. A ball of mass 'm' moving horizontally which velocity 'u' hits a wedge of mass 'M'. The wedge is situated on a smooth horizontal surface. After striking with wedge, the ball starts moving in vertical direction and the wedge starts moving in horizontal plane. Calculate



- the velocity V of wedge.
- the velocity v at which the ball moves in vertical direction.
- the impulse imparted by the ball to the wedge.
- the coefficient of restitution e?

Solution

- Since there is no external force in horizontal direction, momentum in the horizontal direction is conserved.



$$\Rightarrow mu + 0 = 0 + MV \Rightarrow V = mu/M$$

- The velocity of ball along the tangent does not change.

$$\Rightarrow u \cos \theta = v \sin \theta \therefore v = u \cot \theta$$

- The impulse imparted by ball to wedge is equal in magnitude to the impulse imparted by wedge to ball. Its value is

$$mv \cos \theta - (-mu \sin \theta) = m(u \cot \theta) \cos \theta + mu \sin \theta = mu \csc \theta$$

- Along the line of impact, $e = \frac{v_2 - v_1}{u_1 - u_2}$

Here, $u_1 = u \sin \theta$, $u_2 = 0$, $v_1 = -v \cos \theta$, $v_2 = V \sin \theta$

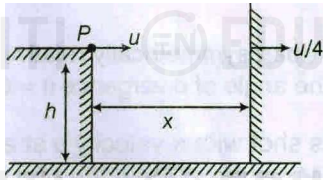
$$\therefore e = \frac{V \sin \theta + v \cos \theta}{u \sin \theta - 0} = \frac{(mu/M) \sin \theta + (u \cot \theta) \times \cos \theta}{u \sin \theta}$$

$$= \boxed{\frac{m}{M} + \cot^2 \theta}$$

PARAGRAPH TYPE

Paragraph for Question Nos. 9 to 10

A particle is thrown with a velocity u from a height h horizontally towards a vertical wall which is moving away with a speed $u/4$ as shown in figure. The particle returns to the point of projection after suffering two elastic collisions (one with the wall and the other with the ground).



9. The total time of flight is
 (a) $2\sqrt{2h/g}$ (b) $2\sqrt{h/g}$
 (c) $\sqrt{2h/g}$ (d) $4\sqrt{2h/g}$
10. What is the initial separation x between the particle and the wall?
 (a) $2u\sqrt{h/2g}$ (b) $u\sqrt{h/2g}$
 (c) $u\sqrt{2h/g}$ (d) $u\sqrt{h/g}$

Solution

- 9.(a) The magnitude of vertical velocity does not change in any collision. The total time of flight is the sum of the time it takes to descend height h and again return to height h .

$$\therefore T = \sqrt{\frac{2h}{g}} + \sqrt{\frac{2h}{g}} = 2\sqrt{\frac{2h}{g}}$$

10. Let v be the horizontal velocity (towards left) of particle after collision. Since collision is elastic, $v_2 - v_1 = u_1 - u_2$

$$\Rightarrow u/4 - (-v) = u - u/4 \quad \Rightarrow v = u/2$$

$$\text{Time taken for collision, } t_1 = \frac{x}{u - u/4} = \frac{4x}{3u}$$

$$\text{In this time, wall has moved by } \frac{u}{4} t_1 = \frac{u}{4} \times \frac{4x}{3u} = \frac{x}{3}$$

Time taken to reach P after collision,

$$t_2 = \frac{x + x/3}{v} = \frac{4x/3}{u/2} = \frac{8x}{3u}$$

Total time, $T = t_1 + t_2$

$$\Rightarrow 2\sqrt{\frac{2h}{g}} = \frac{4x}{3u} + \frac{8x}{3u} = \frac{4x}{u}$$

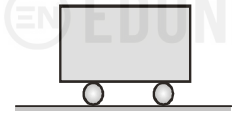
$$\therefore x = u\sqrt{h/2g}$$

Paragraph for Question Nos. 11 to 13

When the mass of a system is variable, a thrust force acts on it in addition to all other forces acting on it. This thrust force is given by $\vec{F} = \vec{v}_r (dm/dt)$ where m is the instantaneous mass of the system and $\vec{v}_r$ is the relative velocity with which the mass dm either enters or leaves the system. A car has total mass 50 kg. Gases are ejected from this backwards with relative velocity 20 m/s.

The rate of ejection of gas is 2 kg/s. Total mass of gas is 20 kg. Coefficient of friction between the car and road

is $\mu = 0.1$ (Take $g = 10 \text{ m/s}^2$)



11. Car will start moving after time
 (a) 4 s (b) 10 s
 (c) 5 s (d) 8 s
12. Maximum speed of car will be (Take $\ln(4/3) = 0.28$)
 (a) 0.6 m/s (b) 0.8 m/s
 (c) 1.0 m/s (d) 1.2 m/s
13. Car will stop after (from starting)
 (a) 12.2 s (b) 6.4 s
 (c) 10.6 s (d) 5.8 s

Solution

11. (c) The car will start moving after the force of thrust becomes equal to the friction force.

Here, $v_r = -20 \text{ m/s}$ and $m = (50 - 2t) \text{ kg}$

$$\Rightarrow \mu mg = v_r (dm/dt)$$

$$\Rightarrow 0.1 \times (50 - 2t) \times 10 = (-20) \times (-2) \quad \therefore t = 5 \text{ s}$$

12. (a) Mass of the gas is 20 kg. At $t = 10 \text{ s}$, all gas would be removed and thrust will stop acting. After $t = 10 \text{ s}$, the car will decelerate due to friction force. So, maximum speed of the car will be at $t = 10 \text{ s}$.

$$\text{Now, } m \frac{dv}{dt} = v_r \frac{dm}{dt} - \mu mg$$

$$\Rightarrow (50 - 2t) \frac{dv}{dt} = (-20) \times (-2) - 0.1 \times (50 - 2t) \times 10$$

$$\Rightarrow \frac{dv}{dt} = \frac{2t - 10}{50 - 2t} = -\left(1 + \frac{20}{t - 25}\right)$$

$$\Rightarrow \int_0^{v_{\max}} dv = -\int_5^{10} \left(1 + \frac{20}{t - 25}\right) dt$$

$$\therefore v_{\max} = 0.6 \text{ m/s}$$

13. (c) Let car stops after time t_0 . After $t = 10 \text{ s}$,

$$\frac{mdv}{dt} = -\mu mg$$

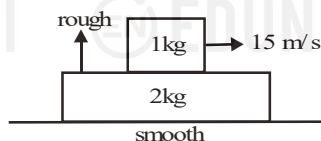
$$\Rightarrow dv = -0.1 \times 10 dt = -dt$$

$$\Rightarrow \int_{v_{\max}}^0 dv = - \int_0^{t_0} dt \Rightarrow v_{\max} = t_0 - 10$$

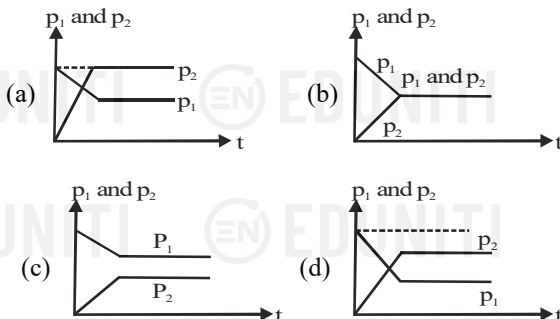
$$\therefore t_0 = 10.6s.$$

Paragraph for Question Nos. 14 to 16

A 1 kg block is given a velocity of 15 m/s towards right over a very long rough plank of mass 2 kg as shown in figure.



14. The correct graph showing linear momentum of 1 kg (i.e. p_1) and of 2 kg (i.e. p_2) versus time is



15. If coefficient of friction between the two blocks is equal to 0.4, then magnitude of initial slope of p_1 versus t and p_2 versus t (in SI unit) will be (Take $g = 10 \text{ m/s}^2$)
 (a) 4 and 2 (b) 2 and 4
 (c) 4 and 4 (d) 2 and 2
16. Momentum of both the blocks is equal at time
 (a) 1.75 s (b) 1.875 s
 (c) 2.5 s (d) 1.25 s

Solution

14. (d) With time, p_1 will decrease and p_2 will increase. After some time both blocks will move together with same velocity. At that time p_2 will be twice of p_1 as mass of lower block is double the mass of upper block.

15. (c) Slope of p vs t graph is force. Its magnitude on both blocks is equal given by

$$\mu m_1 g = 0.4 \times 1 \times 10 = 4N$$

16. (b) $a_1 = \mu m_1 g / m_1 = -\mu g = -0.4 \times 10 = -4 \text{ m/s}^2$

$$a_2 = \mu m_1 g / m_2 = 0.4 \times 1 \times 10 / 2 = 2 \text{ m/s}^2$$

$$p_1 = m_1(u_1 + a_1 t) = 1 \times (15 - 4t)$$

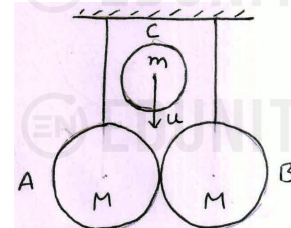
$$p_2 = m_2(u_2 + a_2 t) = 2 \times (0 + 2t) = 4t$$

$$p_1 = p_2 \text{ when } 15 - 4t = 4t \text{ or } t = 1.875 \text{ s}$$

Paragraph for Question Nos. 17 to 19

Two identical balls A and B, each of mass $M = 2 \text{ kg}$ and radius R , are suspended vertically from inextensible strings

as shown in fig. The third ball C of mass $m = 1 \text{ kg}$ and radius $r = (\sqrt{2} - 1)R$ falls and hits A and B symmetrically with speed $u = 10 \text{ m/s}$. Speed of both A and B just after the collision is 3 m/s.



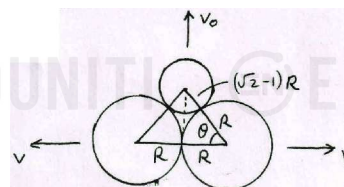
17. Impulse provided by each string during collision is
 (a) $6\sqrt{2} \text{ N s}$ (b) 12 N s
 (c) $3\sqrt{2} \text{ N s}$ (d) 6 N s

18. Speed of C just after collision is
 (a) 2 m/s (b) $2\sqrt{2} \text{ m/s}$
 (c) 5 m/s (d) $(\sqrt{2} - 1) \text{ m/s}$

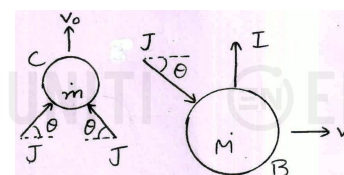
19. The value of coefficient of restitution is
 (a) $1/4$ (b) $1/\sqrt{2}$
 (c) $\sqrt{2} - 1$ (d) $1/2$

Solution

17. (d)



$$\cos \theta = \frac{R}{R + (\sqrt{2} - 1)R} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$



Let the impulse provided to ball B by ball A be J and by string be I . Then, $J \cos \theta = Mv$ and $J \sin \theta = I$

$$\therefore I = Mv \tan \theta = 2 \times 3 \times \tan 45^\circ = 6 \text{ N s.}$$

18. (a) $J = Mv \sec \theta = 2 \times 3 \sec 45^\circ = 6\sqrt{2} \text{ N s.}$

For ball C in vertical direction,

$$2J \sin \theta = m[v_0 - (-u)]$$

$$\Rightarrow 2 \times 6\sqrt{2} \sin 45^\circ = 1 \times (v_0 + 10)$$

$$\therefore v_0 = 2 \text{ m/s.}$$

19. (d) Here, $u_1 = u \sin \theta = \frac{10}{\sqrt{2}} \text{ m/s}$, $u_2 = 0$,

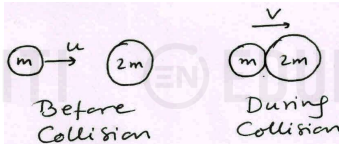
$$v_1 = -v_0 \sin \theta = -\frac{2}{\sqrt{2}} \text{ m/s}, v_2 = v \cos \theta = \frac{3}{\sqrt{2}} \text{ m/s}$$

$$\therefore e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{3/\sqrt{2} - (-2/\sqrt{2})}{10/\sqrt{2} - 0} = \frac{1}{2}$$

MORE THAN ONE CORRECT

20. A particle moving with kinetic energy = 3J makes an elastic head on collision with a stationary particle which has twice its mass. During the impact,
- the minimum kinetic energy of the system is 1J.
 - the maximum elastic potential energy of the system is 2J.
 - momentum and total energy are conserved at every instant.
 - the ratio of kinetic energy to potential energy of the system first decreases and then increases.

20 (a,b,c,d)

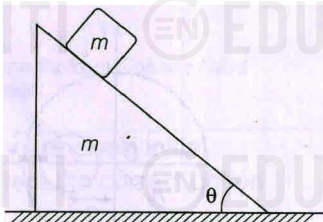


During collision, speed of m first decreases and that of $2m$ increases. When speeds of both becomes equal (say v), KE is maximum and PE is maximum due to maximum deformation. After this stage, speed of m decreases and that of $2m$ increases till the time deformation is completely recovered. Momentum and total energy (KE + PE) are conserved at every instant. Between initial and maximum deformation stage, $mu + 0 = 3mv$

$$\therefore KE_{\min} = \frac{1}{2}(3m)v^2 = \frac{1}{2}(3m)\left(\frac{4u}{3}\right)^2 = \frac{1}{3}\left(\frac{1}{2}mu^2\right) = \frac{1}{3} \times 3 = 1\text{J}$$

$$\therefore PE_{\max} = E_{\text{total}} - KE_{\min} = 3 - 1 = 2\text{J}.$$

21. A block of mass m slides down on an inclined plane of a wedge of mass m (see figure). Friction is absent everywhere. Which of the following statement(s) is/are correct?



- (a) Vertical component of acceleration of block is

$$\frac{g \cos^2 \theta}{1 + \sin^2 \theta} \text{ downwards}$$

- (b) Acceleration of centre of mass of (block + wedge) is

$$\frac{g \sin^2 \theta}{1 + \sin^2 \theta} \text{ downwards}$$

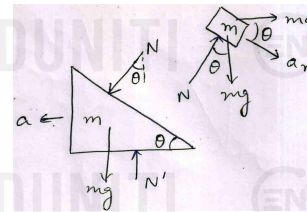
- (c) Acceleration of centre of mass of (block + wedge) is

$$\frac{g \cos \theta}{1 + \sin^2 \theta} \text{ downwards}$$

- (d) Vertical component of acceleration of block is

$$\frac{2g \sin^2 \theta}{1 + \sin^2 \theta} \text{ downwards}$$

- 21 (b, d) Let a be the acceleration of wedge towards left. In the frame of wedge, pseudo force equal to ma acts on block towards right and acceleration (say a_r) is along the incline.



For wedge, along horizontal, $ma = N \sin \theta$

.....(1)

For block, along normal, $N + ma \sin \theta = mg \cos \theta$

.....(2)

and along incline, $ma_r = mg \sin \theta + ma \cos \theta$

.....(3)

From (1) and (2), $a = \frac{g \cos \theta \sin \theta}{1 + \sin^2 \theta}$

On putting this in (3), we get $a_r = \frac{2g \sin \theta}{1 + \sin^2 \theta}$

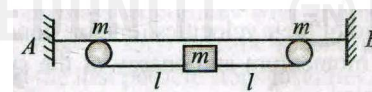
The vertical component of block's acceleration is

$$a_v = a_r \sin \theta = \frac{2g \sin^2 \theta}{1 + \sin^2 \theta} \text{ downwards.}$$

Since, there is no external force on the system in horizontal direction, COM moves only vertically downwards

$$\text{given by } a_{cm} = \frac{ma_v}{m+m} = \frac{g \sin^2 \theta}{1 + \sin^2 \theta}$$

22. Two small rings, each of mass ' m ', are connected to the block of same mass ' m ' through an inextensible massless strings of length l each. Rings are constrained to move over smooth rod AB. Initially, the system is held at rest as shown in fig. Let u and v be the velocities of ring and block, respectively when string makes an angle 60° with the vertical. Then,



(a) $u = \sqrt{\frac{gl}{5}}$

(b) $u = \sqrt{\frac{8gl}{5}}$

(c) $v = \sqrt{3gl}$

(d) $v = \sqrt{\frac{3gl}{5}}$

22. (a, d) The velocities along the string at its two ends is same.

$$\Rightarrow mu \sin 60^\circ = mv \cos 60^\circ$$

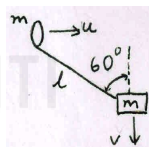
$$\Rightarrow v = \sqrt{3}u$$

Loss in PE = Gain in KE

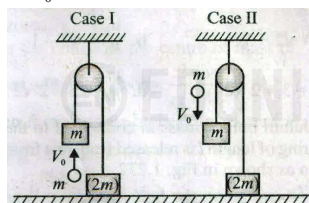
$$\Rightarrow mgl \cos 60^\circ = \frac{1}{2}mv^2 + 2 \times \frac{1}{2}mu^2$$

$$\Rightarrow gl = (\sqrt{3}u)^2 + 2u^2 = 5u^2$$

$$\therefore u = \sqrt{gl/5} \text{ and } v = \sqrt{3}u = \sqrt{3gl/5}$$



23. Two masses $2m$ and m are connected by an inextensible light string. The string is passing over a light frictionless pulley. The mass $2m$ is resting on a surface and mass m is hanging in air as shown in fig. A particle of mass m strikes mass m from below in case (I) and from top in case (II) with a velocity v_0 and sticks to it.



- The conservation of linear momentum can be applied in both the cases just before and just after collision.
- The conservation of linear momentum can be applied in case I but cannot be applied in case II just before and just after collision.
- The ratio of velocities of mass m just after collision in first and second case is $1/2$.
- The ratio of velocities of mass m just after collision in first and second case is 2 .

- 23 (b,d) The momentum will be conserved in first case. In second case, change in momentum will be equal to impulse generated in the string.

In first case, $(m + m)v_1 = mv_0$

$$\Rightarrow v_1 = v_0/2$$

In second case, $(m + m + 2m)v_2 = mv_0$

$$\Rightarrow v_2 = v_0/4$$

$$\text{Hence, the ratio } \frac{v_1}{v_2} = \frac{v_0/2}{v_0/4} = 2$$

24. Object A strikes the stationary object B with a certain given speed u head-on in an elastic collision. The mass of A is fixed, you may choose the mass of B appropriately for following cases. Then after the collision,

- for B to have the greatest speed, choose $m_B = m_A$
- for B to have the greatest momentum, choose $m_B \gg m_A$
- for B to have the greatest speed, choose $m_B \ll m_A$
- for the maximum fraction of kinetic energy transfer, choose $m_B = m_A$

24. (b, c, d)

Using $m_A u = m_A v_A + m_B v_B$ and $v_B - v_A = eu$, we get

$$v_B = \frac{2m_A u}{m_A + m_B} = \frac{2u}{1 + m_B/m_A}$$

v_B is maximum when $m_B \ll m_A$.

$$\text{Momentum, } p_B = m_B \frac{2m_A u}{m_A + m_B} = \frac{2m_A u}{1 + m_A/m_B}$$

p_B is maximum when $m_B \gg m_A$.

Kinetic energy,

$$K_B = \frac{1}{2}m_B \left(\frac{2m_A u}{m_A + m_B} \right)^2 = \frac{1}{2}m_A u^2 \times \frac{4}{\left(\sqrt{m_A/m_B} + \sqrt{m_B/m_A} \right)^2}$$

$$\text{Fraction of KE transfer} = \frac{4}{\left(\sqrt{m_A/m_B} + \sqrt{m_B/m_A} \right)^2}$$

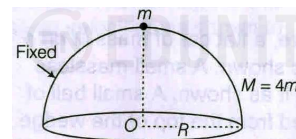
It is maximum when $m_B = m_A$

(Note that for $x > 0$, $x + 1/x$ is minimum when $x = 1$)

25. A small particle of mass m starts sliding down from rest along the smooth surface of a fixed hollow hemisphere of mass $M (= 4m)$. The distance of centre of mass of (particle + hemisphere) from centre O of hemisphere, when the particle separates from the surface

$$\text{of hemisphere is } \frac{\sqrt{69}}{5\alpha} R.$$

Find the value of α



25. (3) Let the particle separates at P.

Then, at P, normal reaction $N = 0$

$$\text{So, } mg \cos \theta = \frac{mv^2}{R}$$

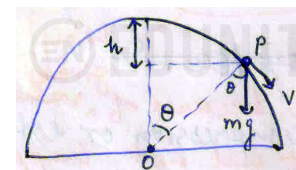
$$\Rightarrow v^2 = Rg \cos \theta \dots (1)$$

$$\text{Also, } \Delta K + \Delta U = 0$$

$$\Rightarrow \frac{1}{2}mv^2 - mgR(1 - \cos \theta) = 0 \dots (2)$$

From, (1) and (2), we get $\cos \theta = 2/3$ and $\sin \theta = \sqrt{5}/3$

If O is origin, the coordinates of P are $\left(\frac{\sqrt{5}}{3}R, \frac{2}{3}R \right)$





Coordinates of COM of hemisphere are $(0, R/2)$

Coordinates of COM of the system are

$$\left(\frac{m\sqrt{5}R/3 + 4m \times 0}{m + 4m}, \frac{m(2R/3) + 4m(R/2)}{m + 4m} \right) = \left(\frac{R}{3\sqrt{5}}, \frac{8R}{15} \right)$$

$$\text{Required distance} = \sqrt{\left(\frac{R}{3\sqrt{5}} \right)^2 + \left(\frac{8R}{15} \right)^2} = \frac{\sqrt{69}R}{15}$$

$$\therefore \alpha = 3$$